

COMPLEX ENGINEERING ACTIVITY (CEA)

ECAT Exam Application — Dual Portal System

Course Name	Fundamentals of Programming & Data Science
Course Code	CMPE-112L
Assignment Type	Complex Engineering Activity (CEA)
Lab / Assignment #	Lab #1
Lab Title	ECAT Exam App — Dual Portal System
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CLO Covered	CLO 2
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1. Problem Statement

Design and implement a Python-based ECAT Exam Application with two separate, credential-protected portals:

- Portal 1 — ECAT Admin Portal: Lets the ECAT team manage the question bank and view all student results.
- Portal 2 — Student Exam Portal: Lets registered students log in, take the exam, and see their result instantly.

Core requirements:

1. Display at least 10 MCQ questions, each with 4 choices (A, B, C, D).
2. Student can answer, skip (S), or type SUBMIT to end early at any time.
3. Exam auto-submits when all questions are answered.
4. Marking Scheme — Correct: +4 marks | Wrong: -1 mark | Skip: 0 marks.
5. Show final score, percentage, grade, and per-question answer review.
6. Admin can view a summary of all student results and drill into any individual result.
7. All data is stored in memory (lists and dictionaries).

2. System Design — Dual Portal Architecture

2.1 Portal Credentials

Portal	Role	Username	Password
Portal 1 — Admin	ECAT Team	ecat_admin	ecat@2024
Portal 2 — Student	Student	student	student123

2.2 Admin Portal Features (8 options)

- Login — 3-attempt logout for security.
- View All Questions — lists every question, its 4 choices, and the correct answer.
- Add New Question — enter subject, question text, 4 choices, correct answer; appended to the in-memory list.
- Delete Question — remove any question by its displayed number.
- Question Bank Statistics — total count and breakdown by subject.
- View All Student Results — summary table showing name, roll, score, percentage, grade, and time for every attempt.
- View Detailed Result (Per Student) — full question-by-question breakdown for any chosen attempt.
- Class Result Statistics — highest, lowest, average score; pass/fail count; grade distribution.

2.3 Student Portal Features (3 options)

- Login — 3-attempt logout; captures full name and roll number for the result record.
- Start Exam — navigate all questions one by one; A/B/C/D to answer, S to skip, SUBMIT to end early.
- View Exam Rules — displays marking scheme and instructions before starting.

3. Data Storage — In-Memory

All data lives in Python lists and dictionaries for the duration of the program run.

Variable	Type	Purpose
questions	List of dicts	Holds all 12 built-in MCQ questions. Admin can add/delete during the session.
all_results	List of dicts	Every student result appended here after exam submission. Admin reads from this list.
answers	Dictionary	Stores {question_index: 'A'/'B'/'C'/'D'/'S'} for the current student's exam.

4. Marking Scheme

Answer Type	Marks	Score Formula
Correct Answer	+4 Marks	correct × 4
Wrong Answer	−1 Mark	wrong × (−1)
Skipped Question	0 Marks	skipped × 0
Total Score		(correct×4) + (wrong×−1)

Grade Boundaries

Grade	Percentage Range	Meaning
EXCELLENT	80% and above	Outstanding performance
GOOD	65% – 79%	Above average performance
AVERAGE	50% – 64%	Satisfactory performance
BELOW AVERAGE	Below 50%	Needs improvement

5. Python Concepts Used in This Program

Every concept below has been covered in CMPE-112L before this CEA.

Concept	Where / How It Is Used
Variables & Data Types	Strings for credentials/names, integers for scores, booleans for login success
Constants	CORRECT_MARKS, WRONG_MARKS, SKIP_MARKS, MIN_QUESTIONS defined at the top
Input / Output	input() for all user entries; print() for menus, questions, results
if / elif / else	Menu navigation, login validation, grade calculation, answer checking
while loop	Main menu loops, login attempt loop, answer validation loop
for loop	Iterating over questions during exam, iterating results for display
Functions	17 functions — every feature is its own function for clean structure
Lists	questions list stores all MCQs; all_results list stores every exam record
Dictionaries	Each question is a dict {id, subject, question, choices, answer}; answers dict maps index→choice
List of Dictionaries	questions and all_results are both lists where every item is a dictionary
String Methods	.strip() cleans whitespace, .upper() normalises A/B/C/D input, .lower() for cancel check
Arithmetic Operators	Score formula: (correct×4)+(wrong×−1); percentage: score/max×100
import time	time.sleep() for brief pauses after login; time.strftime() records exam

	date/time
range() + len()	for idx in range(len(questions)) to iterate questions with an index
.append() / .pop()	.append() adds questions/results to lists; .pop() removes a deleted question
.get() on dict	answers.get(idx, 'S') safely fetches answer or defaults to Skipped
